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Exploring large language models' responses to moral reasoning dilemmas

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ABSTRACT

This study investigates how various large language models (LLMs) generate responses to moral reasoning dilemmas. It specifically examines LLM-generated responses using the Defining Issues Test (DIT-2), which measures abstract moral reasoning schemas, and the Intermediate Concepts Measure (ICM Educational leaders' version), which assesses domain-specific professional moral reasoning. For DIT-2, Claude prioritizes the highest post-conventional moral reasoning, followed by Gemini Advanced and Gemini. For the ICM Educational Leaders version, Gemini Advanced had the highest total ICM score, followed by Claude 3.5 Sonnet and Gemini. The findings indicate that some LLMs can generate responses consistent with sophisticated moral reasoning patterns, producing scores comparable to or exceeding graduate-level human participants; however, no direct comparisons with human participants were made in this study. This study provides a methodological framework for guiding larger-scale research into AI-generated and human moral reasoning patterns.

KEYWORDS

Large language models (LLMs); moral reasoning; Defining Issues Test (DIT-2); Intermediate Concept Measure (ICM)

INTRODUCTION

Artificial intelligence (AI) simulates human intelligence using algorithms, data, and computing power (Haenlein & Kaplan, 2019). The evolution of AI has progressed from simple rule-based systems, including statistical language models, to neural language models and then to advanced deep learning models (Zhao et al., 2023). As an example, in late 2022, OpenAI introduced ChatGPT, a powerful LLM that revolutionized AI applications across industries, excelling in text generation, coding, multilingual processing, and advanced reasoning (Alawida et al., 2023; Bansal et al., 2024; de Winter et al., 2024; Wu et al., 2024; Zhang et al., 2024). LLMs like ChatGPT are built on advanced neural network architectures that use deep learning and natural language processing (NLP). They use advanced NLP, supervised learning, and reinforcement learning techniques to understand and produce text that simulates human conversation (Roumeliotis & Tselikas, 2023). These models have been trained on massive amounts of data, allowing them to respond to a variety of questions in ways that are comparable to humans (Chu et al., 2024; Roumeliotis & Tselikas, 2023; Zubiaga, 2024). In addition, major developments such as large-scale pre-training that captures knowledge from the entire internet, instruction fine-tuning, and Reinforcement Learning from Human Feedback (RLHF) have significantly improved LLM adaptability and performance (Liu et al., 2023).

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The rapid advancement of AI technologies has increased interest in comparing AI systems to humans in various domains, including moral domains (Han, 2023; Sengupta et al., 2025; Tanmay et al., 2023). While the benefits of LLMs in natural language processing and automation are well known, their potential to influence and challenge complex fields such as morality and moral reasoning is opening up new avenues for understanding human ethics, moral alignment, human-AI teaming, and improving decision-making frameworks (Jorgenson et al., 2025; Ramezani & Xu, 2023; Sengupta et al., 2025; Simmons, 2022; Slavkovik, 2022). For example, one study found that LLMs' moral reasoning is influenced by input data, training strategies, and model architectures, which leads to significant variations in their moral judgments (Sachdeva & van Nuenen, 2025). Also, researchers found that LLMs are systematically biased against taking action in moral dilemmas, with a stronger bias than humans. This suggests that blindly relying on LLMs for moral decisions could increase human biases (Cheung et al., 2025). Additionally, LLMs have demonstrated greater decisiveness and structured justifications under certain moral philosophies, while producing ethically controversial outcomes under others (Ding et al., 2025).

LLMs process moral concepts as statistical patterns extracted through language rather than as personally meaningful values developed through lived experience. Their training on large text corpora provides pattern-based recognition of moral concepts common in human societies (Kwon et al., 2023; Yuan et al., 2025).

Modern LLMs incorporate sophisticated safety features and alignment mechanisms, including RLHF and constitutional AI approaches, which are designed to direct outputs toward ethical behavior and away from harmful content (Ganguli et al., 2023). Unaligned LLMs (these models are not programmed with specific human moral values, reflecting diverse societal biases) demonstrate pattern-based recognition of moral concepts because of their exposure to the social totality, but their moral reasoning is still limited and lacks the depth of human understanding (Pock et al., 2023).

The widespread use of LLMs in various industries highlights the importance of understanding their moral reasoning capacities. Since these models are increasingly integrated into decision-making processes in healthcare (Armitage, 2025; Hager et al., 2024), education (Zhui et al., 2024), legal systems (Almeida et al., 2024; Marcos, 2024), art (Ivanova, 2025), and self-driving cars (Ahmad & Takemoto, 2024; Takemoto, 2024; Xu et al., 2025), their ability to generate responses consistent with moral reasoning patterns when navigating ethical complexities is critical. For example, AI systems in healthcare must navigate complex ethical considerations involving patient privacy, diagnostic accuracy, and resource allocation, where modern privacy-preserving techniques such as federated learning and differential privacy demonstrate that these concerns can be addressed without simple trade-offs (Armitage, 2025; Hager et al., 2024). These real-world applications highlight the importance of examining LLMs' functional capabilities to generate responses consistent with moral reasoning patterns and ensuring that they are consistent with human values.

Furthermore, as educational institutions increasingly adopt AI for personalized learning and administrative tasks (Gan et al., 2023; Xing et al., 2025; Yan et al., 2024), understanding how LLMs generate responses to moral dilemmas becomes critical for developing responsible AI systems that promote rather than interfere with moral development. By exploring how different LLMs respond to established moral reasoning tests, this study provides insight into the capabilities and limitations of current AI systems, as well as providing foundations for future research and development in this critical area. As a result, this research aims to determine how LLMs generate responses consistent with moral reasoning patterns across various versions of LLMs to identify differences, similarities, and potential insights into how AI simulates human moral reasoning.

THE IMPORTANCE OF AI IN MORAL REASONING

LLMs can generate responses consistent with moral reasoning patterns using top-down approaches based on normative and descriptive ethical theories, such as normative ethical theories (deontology, utilitarianism) and descriptive frameworks (Theory of Dyadic Morality). Top-down approaches start with predefined ethical principles, rules, or theories, guiding decision-making and moral judgments, while bottom-up

systems develop ethical understanding through experience or crowd-sourced inputs, observing patterns in data. These approaches provide better explainability and structure than traditional bottom-up approaches, which learn ethics from data without explicit moral guidelines (Zhou et al., 2023).

Research shows LLMs can distinguish between moral frameworks (deontology, utilitarianism, virtue ethics) in their representations (Agarwal et al., 2024; Samway et al., 2025). However, as AI systems increasingly engage in human-like interactions, aligning their outputs with human ethical standards has become critical and determining which values to align with remains difficult. Utilitarian frameworks prioritize maximizing overall well-being while struggling to define utility and protect minority interests, whereas the deontological framework (Kantian ethics) prioritizes universal norms and individual dignity but may be too rigid for complex real-world situations (Gabriel, 2020; Mitov, 2021). Recent research has focused on applying virtue ethics to AI agents, but genuine moral character, which emerges from human emotional experience and lived challenges, may be impossible for machines to achieve (Kristjánsson, 2010, 2013; Stenseke, 2023, 2024). In addition, the translation of these principles into computer models poses significant challenges due to the complexity of different situations and the context-sensitivity required for phronesis (practical wisdom) to function properly (Stenseke, 2023, 2024).

Human rights-based frameworks provide a common legal-moral language, but they face difficulties in reaching a global consensus (Hogan & Lasek-Markey, 2024). As a result, addressing AI's ethical concerns requires the combination of concepts from multiple ethical frameworks. This approach facilitates the integration of ideas from various frameworks, resulting in a more comprehensive and effective approach to ethical AI development. This combination allows for addressing the challenges of aligning AI behavior with various human values and ethical considerations (Prem, 2023). While future AI development might require the integration of multiple ethical frameworks to address various moral concerns, the current study focuses on determining how LLMs align with established human moral reasoning patterns. In fact, instead of proposing new AI ethical frameworks, this study explored whether current AI systems have moral reasoning abilities that are consistent with well-validated models of human moral development.

As AI systems become more effectively incorporated into various aspects of daily life, the significance of AI systems that can represent and respond to human values and engage in moral reasoning and judgment increases (Peterson & Gärdenfors, 2024; Sullivan & Fosso Wamba, 2022). The potential challenges of using AI to make ethical decisions have been the subject of recent research (Slavkovik, 2022). For example, AI in autonomous vehicles needs to be able to make nuanced moral decisions in life-or-death circumstances. These decisions include identifying the least harmful options in unavoidable accident scenarios, such as whether to prioritize passenger over pedestrian safety (Awad et al., 2018). The capacity of AI to engage in moral reasoning in these circumstances is both a technical challenge and an ethical responsibility.

Moreover, the cultural specificity of moral norms adds another layer of complexity to AI moral reasoning. What is considered ethical can vary significantly across different cultures and contexts, which creates concern about whose values should be encoded in AI systems. As a result, this challenge is particularly relevant for LLMs trained on diverse datasets that may contain conflicting moral perspectives. Research has shown that English pre-trained Language models (EPLMs) generate a certain level of moral knowledge but are heavily biased toward Western values, making it difficult for them to generalize to diverse cultural contexts (Jinnai, 2024; Ramezani & Xu, 2023).

Another limitation is the possibility of "moral mimicry," in which LLMs generate moral rationalizations that mimic human reasoning without engaging with the complex moral implications involved in a situation. For example, research found that LLMs can produce moral biases related to political identities in morally loaded responses to political prompts, which indicates that these models simulate patterns from their training data (Simmons, 2022).

Despite these limitations, exploring LLMs' responses to standardized moral reasoning tests provides useful information about their capabilities and potential applications. In addition, through comparing different models across various measures and examining the patterns in their answers,

researchers can obtain a deeper understanding of how these systems process and generate moral reasoning, find areas for improvement, and develop more ethically aligned AI systems.

Previous research on human participants has demonstrated that sophisticated moral reasoning is necessary for societal functioning and personal development. Higher levels of moral reasoning have been linked to prosocial behavior, ethical decision-making in professional settings, and the ability to solve complex moral dilemmas (Kohlberg & Power, 1981; Rest et al., 2000). In democratic societies, the capacity to engage in post-conventional moral reasoning which prioritizes universal ethical principles over mere rule-following enables critical evaluation of social norms and institutions, which results in contributing to social progress and justice. For instance, individuals with advanced moral reasoning abilities can make decisions that balance competing ethical principles while also considering multiple stakeholders, which is increasingly important in today's complex, pluralistic world.

Creating AI systems with sophisticated moral reasoning response capabilities is important as they become more integrated into decision-making processes across multiple domains. As a result, by exploring how various LLMs respond to moral reasoning evaluations, researchers can gain a better understanding of their capabilities and limitations, identify areas for improvement, and create more ethically aligned AI systems capable of navigating complex moral dilemmas in ways that are consistent with human values.

A recent study used the Behavioral Defining Issues Test to assess ChatGPT's moral reasoning capacity. The findings revealed that ChatGPT provided moral reasoning responses equal to human participants, scoring 45.83 in post-conventional reasoning (slightly lower than the median score of 50.00 achieved by human undergraduate students, with a trivial effect size difference). These findings indicate that LLMs such as ChatGPT have the potential to generate responses that functionally resemble features of human moral reasoning (Han, 2023). In a comparative study of moral decision-making, narratives generated by human participants and ChatGPT-3 were assessed using metrics such as causality, explicability, and overall satisfaction. The study showed no significant difference in the quality of explanations between human and AI-generated responses, which indicates that ChatGPT-3 can generate moral justifications comparable to those of humans (Rehman et al., 2025).

Another group of researchers used the Defining Issues Test (DIT-1) to assess the moral reasoning abilities of LLMs, such as GPT-3, GPT-3.5, GPT-4, ChatGPT (July 2023), Llama2-Chat, and PaLM-2. The results varied among models; for example, GPT-3 performed closest to a random baseline, indicating inadequate moral reasoning abilities. However, GPT-4 performed well, with a P-score of 55.68, indicating a post-conventional level of moral reasoning equivalent to that of graduate students (advanced level explained further below). In addition, ChatGPT, Llama2-Chat, and PaLM-2 showed lower reasoning abilities than young adults or college students. The study emphasized GPT-4's significant improvements over prior versions but also found inconsistency across various scenarios. Researchers proposed that the models' reasoning abilities were the result of extensive training data and improved Reinforcement Learning from Human Feedback (RLHF) (Tanmay et al., 2023).

Studies on LLMs trained with RLHF demonstrate that models can avoid harmful outputs such as bias and discrimination when given clear ethical instructions. This ability to generate responses that align with ethical guidance through natural language instructions is a positive sign for the future of LLMs in moral reasoning tasks (Askell et al., 2021; Ganguli et al., 2022, 2023; Gehman et al., 2020). In essence, LLMs can be trained to produce responses that follow moral reasoning protocols by using their capacity to internalize ethical instructions learned through human feedback (Ganguli et al., 2023).

This area of study continues to grow, and there are challenges such as enabling AI systems to perform moral reasoning, address bias, and ensure transparency (Slavkovik, 2022). This concern highlights the need for additional development of LLMs to better align with human moral frameworks (Pock et al., 2023; Slavkovik, 2022). This study addresses this gap by exploring the generative responses of various LLMs to well-known moral reasoning tools such as the Defining Issues Test (DIT-2) and Intermediate Concept Measure (ICM) (Kerr, 2018; S. J. Thoma & Dong, 2014). The present article

explores the degree to which LLMs generate responses that functionally resemble human moral reasoning and provides an extensive evaluation of the compatibility of various LLM versions with human moral reasoning frameworks.

NEO-KOHLBERGIAN THEORY OF HUMAN MORAL FUNCTIONING

The study reported in this article explores LLMs' responses to moral reasoning dilemmas through the lens of neo-Kohlbergian theory, including the Four-Component Model (FCM) (Rest, Thoma, et al., 1999). FCM explains morality through four psychological processes (Narvaez & Rest, 1995, 1994). First, Moral sensitivity involves detecting potential harm to others, being concerned about others' welfare, and recognizing ethical issues. Second, moral judgment determines right and wrong actions and provides rationales for behavioral decisions. Moral reasoning involves weighing up the moral features of a situation to decide how to respond. Third, moral motivation prioritizes moral values over personal gains or social approval as reasons for one's actions, and finally, moral character involves acting on moral values and judgments. Each component of the FCM plays a distinct but interconnected role in moral functioning (Narvez & Rest, 1994).

The focus of the reported study is on the moral reasoning component, where most neo-Kohlbergian expertise can be found. Moral reasoning is the cognitive process that happens when individuals make decisions about what is morally right or wrong in a particular situation (Kohlberg & Power, 1981; Richardson, 2003); for example, a father contemplates stealing food for his starving family from the warehouse of a rich man hoarding food (S. J. Thoma & Dong, 2014).

Within the component of moral judgment, there are three levels: codes of conduct, intermediate concepts, and bedrock schemas. Codes of conduct are specific rules requiring minimal interpretation; intermediate concepts involve virtue-like concepts involving contextualized understanding, such as concepts of informed consent or virtues such as honesty and courage. Intermediate concepts evaluated by Intermediate Concept Measures (ICMs) that need to be developed uniquely for a context, such as a profession (teaching) or stage of development (adolescent). ICMs are, therefore, uniquely focused on practical contextualized moral reasoning (Thoma et al., 2013).

Bedrock moral schemas are considered global schemas involving a person's general moral orientation to the world. Developed from Kohlberg's original stage-based theory, moral schemas are evaluated by the Defining Issues Test (DIT). The DIT measures moral reasoning and activates three moral schemas: personal interests, maintaining norms, and post-conventional (Rest, Narvaez, Bebeau & Thoma, 1999; Rest et al., 2000). The neo-Kohlbergian approach underpinning this perspective places particular emphasis on progression to a post-conventional level of moral reasoning in adolescents and adults as the highest level of moral reasoning. Dropping Kohlberg's original stage-based approach for moral schemas allows for fluidity between "levels" such that individuals may use different schemas depending on the situation, which leads to a more profound understanding of moral development (S. J. Thoma & Dong, 2014).

At the lowest level of moral reasoning, the personal interest schema involves individuals motivated primarily by self-interest, individual benefits, and social approval rather than broader ethical principles (Rest, Narvaez, Bebeau, et al., 1999). For LLMs, this schema shows in response patterns that highlight individual rewards, avoid negative consequences, or seek approval, as learned from training data. Low personal interest scores in advanced models show successful alignment with universal ethical principles rather than self-centered justifications.

The maintaining norms schema emphasizes the importance of established rules, roles, and formal organizational structures. More advanced than personal interests but still limited, it focuses on maintaining the current social order and adhering to established procedures and authorities. The post-conventional schema, as the most advanced form of moral reasoning, involves nuanced and sophisticated moral reasoning capable of exceeding local norms if they curtail ethical goods. The DIT P-score represents the likelihood of utilizing the postconventional schema, focusing on universal ideals and full reciprocity of social norms, which emphasizes principles that can be justified in a society

committed to moral values. Overall, post-conventional reasoning includes a critical examination of societal norms as well as a commitment to universal principles of justice, rights, and welfare (Rest, Narvaez, Bebeau, et al., 1999; S. Thoma, 2014).

Neo-Kohlbergian theory, involving bedrock moral schemas and intermediate concepts, provides a framework for assessing and potentially improving LLM moral reasoning abilities, which is useful for AI development. In an ideal world, advanced AI systems would be able to engage in post-conventional reasoning, balancing universal ethical principles with contextual awareness. However, achieving this level of moral reasoning in AI systems presents some challenges, particularly given the complexity and sometimes contradictory nature of human moral reasoning as well as cultural differences. Researchers can gain a better understanding of specific AI's current capabilities and identify areas for improvement in future AI development by examining how different LLMs perform on measures such as DIT-2 (as the latest version of DIT) and ICM.

METHOD

The study used multiple LLM platforms alongside two moral reasoning measures (DIT-2 and ICM for Educational Leaders). The LLM models included in the research are: ChatGPT-3.5, ChatGPT-4, ChatGPT-4O, Grok Premium Plus, Claude 3.5 Sonnet, Gemini, and Gemini Advanced. These LLMs were chosen based on their popularity and recognition as of summer 2024, ensuring the study examined the most widely used models. The diversity of these models, with their unique architectures, training techniques, and design principles, provides an adequate sample for assessing moral reasoning abilities.

MORAL REASONING TOOLS

DIT-2 and ICM use recognition tasks in which participants rate and rank pre-formulated moral considerations. DIT-2 assesses bedrock schemas (reasoning about societal institutions and universal principles), whereas ICM focuses on contextualized, domain-specific moral reasoning in professional settings.

DEFINING ISSUES TEST (DIT2)

The Defining Issues Test (DIT) is a widely used assessment tool for measuring moral reasoning. DIT-2 is an updated version of the original DIT, incorporating more contemporary moral dilemmas and improved scoring algorithms. The DIT has been extensively validated and is supported by many studies, which shows its strong reliability and validity (Gungordu et al., 2024; S. J. Thoma & Dong, 2014).

DIT-2 includes several indices that help measure other aspects of moral reasoning. These indices are: The Type Indicator (TYPENEW): the dominant moral schema is represented by the Type Indicator, which ranges from 1 to 7. Type 7 indicates the predominant postconventional schema that is consolidated. In contrast, Type 4 represents predominant maintaining norms schema that is consolidated. In addition, Types 1, 4, and 7 are consolidated profiles, while types 2, 3, 5, and 6 are transitional profiles. Utilizer Score (U): The Utilizer Score (−1 to 1) measures consistency between moral reasoning and action choices. Scores closer to 1 indicate greater agreement between item endorsement and action choice, whereas scores closer to −1 indicate less agreement. Humanitarian Liberalism (HUMLIB): This score (0–5) is based on the number of humanitarian liberal action choices made. Higher scores indicate stronger alignment with positions endorsed by experts in political science and philosophy. Cannot Decide Choices (NUMCD): This indicates how many times "can't decide" was chosen across the dilemmas (1–5). Indecision can be linked to developmental stages, with transitional phases frequently showing increased indecision due to multiple conflicting interpretations. Religious Orthodoxy (CANCER10): This is a proxy measure (1–9) based on item 10 in story 4

(the cancer story). Lower scores indicate less religious orthodoxy. It is calculated as the sum of the rating and weighted rank given to an item that evokes the notion that only God should determine whether someone lives or dies. Consolidation/Transition (CONSTRAN): A score of 2 indicates a consolidated profile, meaning there is clear evidence of schema preference. A score of 1 would indicate a transitional profile, reflecting developmental disequilibrium with less discrimination among schema-typed items (Bebeau & Thoma, 2003).

The four primary indices which represent the likelihood of utilizing each schema are the most important indices for the DIT2. As mentioned earlier, the Personal Interest Schema score (0–100) is the lowest level of moral reasoning. The Maintaining Norms schema score (0–98) represents maintaining societal norms, including the existing legal system, existing roles and formal organizational structure, whereas the post-conventional moral schema score (0–95) shows sophisticated moral reasoning. The P-Score is an overall index of schema consideration, and is the traditional score used in most DIT research. Finally, the N2 score (0–95) is a relatively new indicator that assesses both the prioritization of post-conventional items and the de-emphasis of personal interest items. N2 score provides a more nuanced understanding of moral development than just the P score (Bebeau & Thoma, 2003).

There are some additional Indices. The Antisocial Score (ASCORE): it ranges from 0 to 16, with higher scores indicating stronger antisocial/anti-establishment attitudes. These considerations assume understanding of stage 4, but criticize existing authorities for hypocrisy (also referred to as stage 4½). Meaningless Items (MSCORE): Scales from 0 to 16, indicating endorsement of items with complex but meaningless wording. Higher scores may indicate attempts to fake higher scores and/or a lack of understanding. Participants who receive scores greater than 10 will be purged. New Checks (TOTCC): A reliability check that combines multiple indicators to detect invalid responses. Participants who receive scores greater than 200 will be purged. It checks for rate-and-rank consistency, missing data, endorsement of meaningless items, and nondiscrimination (giving the same rating to too many items). Purged Participants (PURGED): A binary indicator (1 = Included, 0 = Purged) that indicates whether a respondent's data was retained or excluded from analysis due to reliability checks (Bebeau & Thoma, 2003).

The DIT-2 comprises five moral dilemmas: (1) Famine: a father contemplates stealing food for his starving family from the warehouse of a rich man hoarding food; (2) Reporter: a newspaper reporter must decide whether to report a damaging story about a political candidate; (3) School Board: school board chair must decide whether to hold a contentious and dangerous open meeting; (4) Cancer: a doctor must decide whether to give an overdose of pain-killer to a suffering but frail patient; and (5) Demonstration: college students demonstrate against U.S. foreign policy. For each dilemma, participants must decide what the main character should do, rate the importance of 12 considerations, and rank the four most important. This structure allows the DIT-2 to assess participants' preferred actions and, more importantly, their reasoning behind these preferences. The DIT-2 test reveals participants' underlying moral schemas and reasoning patterns by asking them which considerations they prioritize. The scoring procedure involves identifying which schema each consideration represents (personal interest, maintaining norms, or postconventional) and calculating the proportion of responses affiliated with each schema. As mentioned before, DIT-2 includes several reliability checks to ensure the validity of responses. These include identifying participants who choose meaningless items, those who have inconsistent ratings and rankings, and those who are unable to discriminate between items. These checks improve test reliability by ensuring that the data reflect valid moral reasoning rather than random responses or task misunderstandings (Bebeau & Thoma, 2003).

When applied to LLMs, DIT-2 provides a standardized framework for evaluating LLMs capabilities to generate responses consistent with moral reasoning patterns. By examining which considerations the LLM models prioritize across different dilemmas, researchers can identify patterns in their moral reasoning and compare them to human developmental patterns. This approach provides useful information about the models' ability to simulate different levels of moral reasoning and their alignment with human moral development.

ICM FOR EDUCATIONAL LEADERS

ICM for Educational Leaders is a recently created ICM that assesses the application of virtue-like concepts to scenarios in educational settings aimed at leaders, mostly principals. The measure consists of four dilemmas representing realistic professional scenarios a school leader might face. After each dilemma is described in the measure, there is a list of potential action choices about what the protagonist in the story might do, followed by a list of reason choices about why the protagonist might act. The ICM scoring method is based on a predetermined expert panel process involving groups of educational leaders. The expert panels worked to agree a scoring key that determines which items from the lists of action and reason choices are adequate, inadequate or neutral. Participants are asked to identify three *best* and *worst* items for both action and reason options. Selecting an adequate score as *best* generates a positive score but selecting an inadequate option as *best* will produce a negative score. Overall, higher scores across the measure in the form of percentage match indicate closer agreement with the expert's assessment of acceptable and unacceptable responses (Bebeau & Thoma, 1999; Kerr, 2018). A score across the entire measure of 0.80 represents an 80% match to the expert panel key. There is no specific right or wrong answer with a range of ways to score well or poorly reflecting real professional life. Being new, this measure is still being tested, and data so far suggest it is somewhat easier to complete than other ICMs.

The ICM for Educational Leaders evaluates how educational leaders apply virtue-like concepts such as fairness, integrity, and responsibility to specific professional scenarios. As mentioned earlier, this measure is pitched at an intermediate level of moral reasoning introduced in response to criticisms that bedrock moral schemas, as assessed by DIT, were too abstract and unrelated to realistic life events. As such, intermediate concepts are a level of moral reasoning situated between abstract moral principles (assessed by DIT-2) and codes of conduct, enhancing understanding of moral reasoning in professional settings (Bebeau & Thoma, 1999). The four dilemmas are as follows: (1) A principal dealing with a high-achieving student suspected of drug use before an important school event, balancing strict drug policies against potential severe consequences for the student; (2) A principal navigating parental requests to move a student with learning disabilities from special education to mainstream classrooms, despite teacher resistance and professional recommendations against it; (3) A principal handling a situation where a teacher's personal iPad has gone missing after a tutoring session with an at-risk student; and (4) A principal addressing cyberbullying conducted partly through school-issued laptops, where a female student is being harassed online by classmates after a relationship ended. Each scenario presents a complex ethical dilemma that necessitates educational leaders to evaluate competing values, including student welfare, policy adherence, professional judgment, parental wishes, faculty concerns, and appropriate outcomes (Center for the Study of Ethical Development, 2025).

The scoring process assesses both action choices (what should be done) and justifications (why it should be done), reflecting the recognition that moral reasoning includes both determining appropriate actions and providing principled justifications for those actions. This dual focus enables the measure to evaluate both the practical and theoretical aspects of moral reasoning in professional settings. The measure is also capable of assessing participants' ability to identify *best* and *worst* actions and reasoning, which, according to other sample groups, appears to be a different skill (Center for the Study of Ethical Development, 2025; Thoma et al., 2019; Walker et al., 2021).

The ICM, therefore, offers a different approach for evaluating LLMs because it reveals their ability to navigate domain-specific moral challenges, balance competing values, and provide principled justifications for their recommendations. This is especially important for understanding how LLMs might function in professional settings where moral reasoning must be applied to specific, contextualized problems rather than abstract principles.

DATA COLLECTION & SCORING PROCEDURE

The study used a systematic methodology including multiple stages. First, DIT-2 and ICM moral reasoning dilemmas were presented to the various LLMs to assess their moral reasoning capacities. Each LLM was presented with standardized prompts containing the moral dilemmas, ensuring consistency across all platforms. For both DIT-2 and ICM, one-shot prompts (which provide a single example to guide the model) were used to replicate instructions provided to human participants. Data collection was conducted in June 2024 to maintain temporal consistency across model responses. The generated responses were then collected and scored using the identified scoring methods for each measure. For the DIT-2, each LLM's responses were scored using the standard scoring procedures established by Rest and colleagues (Rest, Narvaez, Thoma, et al., 1999). This included determining which items the model considered most important for each dilemma, determining which schema each item represented, and calculating the proportion of responses related to each schema. As mentioned earlier, the P score represents the proportion of post-conventional considerations ranked as important, whereas the N2 score includes the post-conventional item prioritization and personal interest item rejection (Bebeau & Thoma, 2003). For the ICM, scoring included comparing the LLMs' selection of *best* and *worst* actions and reasons to expert panel judgment represented by the scoring key in SPSS syntax. Best Actions (ACTBRK) assesses the model's ability to identify appropriate actions across all four dilemmas; Best Reasons (REASBRK) assesses recognition of sound justifications; Worst Actions (ACTWRK) assesses identification of inappropriate behaviors; and Worst Reasons (REASWRK) assesses the ability to recognize flawed justifications. The Overall Worst Ranks (TOTWORSTRK) score reflects the model's ability to identify inappropriate actions and reasons, whereas the Overall Best Ranks (TOTBESTRK) score reflects its ability to identify suitable ones. The total ICM (TOTICM) score combines these two dimensions to provide an overall assessment of alignment with expert judgment (Center for the Study of Ethical Development, 2025).

The Defining Issues Test (DIT-2) is a copyrighted measure that is scored exclusively by the University of Alabama's Center for the Study of Ethical Development. This also applies to the ICM Educational Leadership version, although it is not copyrighted. The scoring algorithm and keys for the DIT-2 and ICM Educational Leadership are confidential and held at the Center for the Study of Ethical Development – these are not publicly available since this would compromise the measures. For more information about the DIT-2 and ICM measures, please see the recently published norm for the measure and the Center for the Study of Ethical Development (Center for the Study of Ethical Development, 2025; Gungordu et al., 2024).

The scoring process involves systematically mapping each LLM's action choice, ranked and rated item selections to their corresponding moral schemas (personal interest, maintaining norms, or post-conventional) using the standardized scoring algorithm. For DIT-2, the LLM's ratings and rankings of these items are quantified to calculate the proportion of responses associated with each schema, which provides the P-score, N2 score, and other indices. Additionally, for ICM, the LLM's selections of best and worst actions and reasons are compared against expert panel judgments encoded in the scoring key to calculate agreement levels. All data were returned to the University of Alabama's Center for the Study of Ethical Development for scoring, which manages the scoring of moral reasoning assessments and provides scoring results for each measure.

RESULTS

Defining Issues Test (DIT-2) Results

The results of the Defining Issues Test (DIT-2) showed differences in moral reasoning abilities among LLMs. Claude 3.5 Sonnet generated responses with the highest level of post-conventional moral reasoning, with a P-score of 72, indicating a strong ability to replicate high-level moral reasoning. Gemini Advanced and Gemini had P-scores of 64 and 58, respectively. These scores are higher than the human average range, which typically falls between 31.1–37.2 for undergraduates and 38.5–42.3

for graduate-level individuals, according to the DIT-2 norms study (Gungordu et al., 2024), as well as those reported in Han's study (45.83) (Han, 2023) and Tanmay's study (55.68) (Tanmay et al., 2023). Other LLM platforms with lower levels of post-conventional reasoning included Grok (48), ChatGPT-4O (44), ChatGPT-4 (44), and ChatGPT-3.5 (18).

The highest N2 score, which emphasizes the presence of post-conventional reasoning and the absence of personal interests, was 71.10 for Claude 3.5 Sonnet. This was followed by Gemini Advanced (60.31) and Gemini (52.11). The remaining scores were ChatGPT-4O (55.07), Grok (47.98), ChatGPT-4 (46.53), and ChatGPT-3.5 (36.20). Additionally, ChatGPT-4O had a higher N2 score (55.07) than P-score (44), indicating that it rejected personal interests more effectively than it prioritized post-conventional reasoning.

A more in-depth analysis of the schema scores reveals intriguing patterns across the LLMs. Gemini Advanced had the highest personal interest schema score (24.00), followed by ChatGPT-3.5, Gemini, and Grok (all at 22.00), then both ChatGPT-4 and ChatGPT-4O (14.00), followed by Claude 3.5 Sonnet (8.00).

ChatGPT-3.5 scored the highest score (60.00) on the maintaining norms schema, indicating a strict adherence to established rules and conventions. ChatGPT-4 and ChatGPT-4O scored the same (42.00), followed by Grok (30.00), Claude 3.5 Sonnet, and Gemini (20.00), with Gemini Advanced scoring the lowest (10.00).

The distribution of schema scores among LLMs shows unique differences in moral reasoning approaches. Table 1 shows these distributions, which emphasize the contrast between post-conventional and N2 scores and the other schemas.

Table 1. Schema distribution and N2 scores across LLMs.

LLMs model	Personal Interest (score range: 0–100)	Maintain Norms (score range: 0–98)	Post Conventional (score range: 0–95)	N2 score (score range: 0–95)
ChatGPT 3.5	22.00	60.00	18.00	36.20
ChatGPT 4	14.00	42.00	44.00	46.53
ChatGPT 4O	14.00	42.00	44.00	55.07
Claude 3.5 Sonnet	8.00	20.00	72.00	71.10
Gemini	22.00	20.00	58.00	52.11
Gemini Advanced	24.00	10.00	64.00	60.31
Grok	22.00	30.00	48.00	47.98

This distribution shows a clear progression from earlier to more advanced LLM models, with newer models typically scoring higher on post-conventional. The most noticeable difference is between ChatGPT-3.5 and Claude 3.5 Sonnet, which has a 54-point difference in P-scores, indicating significant advances in the ability to simulate complex moral reasoning.

The N2 scores, which provide a more sophisticated evaluation of moral reasoning by considering consideration of both the prioritization of post-conventional items and the rejection of personal interest items, show a similar pattern to the P-scores but with some notable differences.

In particular, ChatGPT-4O and ChatGPT-3.5 have higher N2 scores than their P-scores, indicating that, while they may not prioritize post-conventional considerations as strongly as other models, they successfully reject personal interest considerations. This pattern indicates a moral reasoning approach that, while not yet fully developed into post-conventional reasoning, has progressed beyond purely self-centered reasoning.

The overall pattern across all measures indicates significant variation in moral reasoning approaches among LLMs, with newer models showing more advanced moral reasoning abilities. The mean post-conventional score across all LLMs was 49.71 (SD = 17.53), while the mean N2 score was 52.76 (SD = 11.07), which indicates that these LLMs exhibit post-conventional reasoning response patterns at levels comparable to or exceeding those found in typical human adult samples.

These profiles show interesting patterns. Claude 3.5 Sonnet exhibits the strongest orientation toward post-conventional moral reasoning, with little reliance on personal interest schemas, implying a moral reasoning approach based on universal ethical principles rather than self-interest or rigid rule-following. Gemini Advanced exhibits a distinct pattern, with relatively high personal interest scores, high post-conventional scores, and low maintaining norms scores, possibly indicating a moral reasoning approach that balances individual concerns with universal principles while placing less emphasis on established rules and conventions. ChatGPT-4 and ChatGPT-4O have identical schema distributions, implying that moral reasoning approaches remain consistent across these versions despite differences in underlying architectures. Both models exhibit a balance of maintaining norms and post-conventional schemas with low personal interest scores, indicating a moral reasoning approach that values both established norms and universal principles. ChatGPT-3.5 has a significantly different profile, with a strong emphasis on maintaining norms and minimal post-conventional reasoning. This indicates a moral reasoning approach that is heavily reliant on following established rules and conventions rather than critically examining them with universal principles. In addition, this emphasis on rule-following rather than principle-based reasoning stands in contrast to more advanced models like Claude 3.5 Sonnet and Gemini Advanced.

TYPE INDICATORS AND ADDITIONAL INDICES

The Type indicator scores show the dominant moral schema used by each LLM. Most models have been identified as Type 7 (ChatGPT-4, ChatGPT-4O, Claude 3.5 Sonnet, Gemini, Gemini Advanced, and Grok), indicating a consolidated post-conventional schema. Only ChatGPT-3.5 was identified as Type 4, indicating that it primarily used the maintained norms schema, as evidenced by its high maintaining norms score (60.00) and low post-conventional score (18.00).

The Utilizer scores, which assess how closely action choices match moral judgments, differed between models. Gemini Advanced had the most positive Utilizer score (0.31), followed by ChatGPT-3.5 (0.08) and Claude 3.5 Sonnet (0.05). The other models had negative Utilizer scores, ChatGPT-4 (−0.07), Gemini (−0.08), Grok (−0.01), and ChatGPT-4O (−0.13). These scores indicate different levels of consistency between moral reasoning and action choices. In addition, all LLMs had a consolidation transition score of 2.00, indicating comparable levels of schema stability across moral reasoning tasks.

The experimental indices provided more context. ChatGPT-4O and Gemini Advanced scored the highest Humanitarian Liberalism scores (5.00 each), followed by ChatGPT-4 and Grok (both 4.00), Claude 3.5 Sonnet and Gemini (both 3.00), and ChatGPT-3.5 (0.00). Moreover, for Religious Orthodoxy (proxy measure), ChatGPT-4 and ChatGPT-4O scored highest (9.00), followed by ChatGPT-3.5 (6.00), Claude 3.5 Sonnet and Gemini Advanced (both 2.00), and Gemini and Grok scoring lowest (1.00). The results of additional DIT2 indices across the various LLMs are shown in Table 2.

Table 2. Additional DIT-2 indices across LLMs.

LLM Model	Type Indicator	Utilizer Score	Humanitarian Liberalism	Cannot Decide Choices	Religious Orthodoxy
ChatGPT-3.5	4	0.08	0.00	2.00	6.00
ChatGPT-4	7	−0.07	4.00	1.00	9.00
ChatGPT-4O	7	−0.13	5.00	0.00	9.00
Claude 3.5 Sonnet	7	0.05	3.00	2.00	2.00
Gemini	7	−0.08	3.00	1.00	1.00
Gemini Advanced	7	0.31	5.00	0.00	2.00
Grok	7	−0.01	4.00	1.00	1.00
Mean	**6.57**	**0.02**	**3.43**	**1.00**	**4.29**
SD	**1.13**	**0.15**	**1.72**	**0.82**	**3.64**

ICM EDUCATIONAL LEADERS RESULTS

For the ICM Educational Leaders version, several LLMs showed high levels of performance, even considering that ICM Educational Leaders seems to generate higher scores than other ICMs according to human samples tested so far. Gemini Advanced scored the highest total ICM score of 0.90, which indicates a strong alignment with expert evaluations of moral reasoning. Claude 3.5 Sonnet and Gemini both followed closely with scores of 0.86. ChatGPT-4O and ChatGPT-4 both achieved a total ICM score of 0.78, compatible with good human scores in existing samples, while Grok scored 0.61. ChatGPT-3.5 was significantly behind, with a total ICM score of 0.32, which is extremely low. These scores indicate the LLMs' ability to produce strong and consistent responses at the intermediate concept level. The results of ICM total scores and component scores across LLMs are shown in Table 3.

Table 3. ICM total scores and component scores across LLMs.

LLMs model	Best Actions (ACTBRK)	Best Reasons (REASBRK)	Worst Actions (ACTWRK)	Worst Reasons (REASWRK)	Overall Worst Ranks, (TOTWORSTRK)	Overall Best Ranks (TOTBESTRK)	Total ICM (TOTICM)
Gemini Advanced	0.88	0.92	0.83	1.00	0.92	0.90	0.90
Claude 3.5 Sonnet	0.88	0.75	0.92	1.00	0.96	0.81	0.86
Gemini	0.75	1.00	0.75	0.92	0.83	0.88	0.86
ChatGPT-4O	0.71	0.71	0.92	0.92	0.92	0.71	0.78
ChatGPT-4	0.71	0.67	0.92	1.00	0.96	0.69	0.78
Grok	0.54	0.50	0.83	0.75	0.79	0.52	0.61
ChatGPT-3.5	0.25	0.54	0.17	0.17	0.17	0.40	0.32

Note: Score range: −1 to 1.

Analysis of the ICM component indices shows interesting patterns in LLM moral reasoning abilities. For Best Actions (ACTBRK), Gemini Advanced and Claude 3.5 Sonnet had the highest scores (0.88), while ChatGPT-3.5 had the lowest (0.25). On the other hand, for Best Reasons (REASBRK), Gemini earned a perfect score (1.00), followed by Gemini Advanced (0.92), which shows these models' superior ability to identify sound ethical rationales. Claude 3.5 Sonnet, ChatGPT-4, and ChatGPT-4O all achieved 0.92 on Worst Actions (ACTWRK), while ChatGPT-3.5 struggled (0.17). For Worst Reasons (REASWRK), Claude 3.5 Sonnet, ChatGPT-4, and Gemini Advanced all received perfect scores (1.00), demonstrating a remarkable ability to detect flawed reasoning. However, most advanced models performed better on reasoning components (REASBRK and REASWRK) than action components (ACTBRK and ACTWRK), implying that they are better at evaluating justifications than determining appropriate behaviors.

A breakdown of ICM total scores reveals additional details. The "TOTWORSTRK" score, which measures the model's ability to identify the worst choices, shows that ChatGPT-4 performed best (0.96), followed by Claude 3.5 Sonnet, while ChatGPT-3.5 performed poorly (0.17). Gemini Advanced scored the highest (0.90) on the "TOTBESTRK" score, which measures the ability to identify the best options, while ChatGPT-3.5 scored the lowest (0.40).

This result highlights the differences in performance between the two ICM components. TOTWORSTRK scores are high in all advanced models (except ChatGPT-3.5), which indicates that these models are particularly effective at identifying inappropriate actions and justifications in educational leadership contexts. Indeed, this ability to recognize what not to do may be easier for models to learn than identifying optimal solutions, similar to a pattern observed in human moral development in which recognizing clear violations occurs before identifying ideal responses.

In addition, TOTBESTRK scores varied significantly, with Gemini Advanced and Gemini scoring extremely high (0.90 and 0.88, respectively), whereas other models, such as ChatGPT-4 and Grok, scored moderately to low (0.69 and 0.52). This implies that identifying optimal actions and

justifications in complex educational leadership scenarios requires more sophisticated reasoning skills, which are not equally developed across all models. The significant gap between ChatGPT-3.5 and all other models is especially evident in the TOTWORSTRK score, where ChatGPT-3.5 scores only 0.17, compared to Grok's next lowest score of 0.79. This indicates a qualitative difference in the ability to recognize inappropriate actions between earlier and more recent model revisions, which emphasizes the significant advances in capabilities to generate responses consistent with moral reasoning patterns in newer LLMs.

Responses to specific dilemmas provide additional insights. For the "Allison's Challenge" dilemma, which involves maintaining fairness and accountability in school policy, advanced models such as Claude 3.5 Sonnet and Gemini Advanced demonstrated nuanced reasoning that balanced accountability with developmental considerations, whereas ChatGPT-3.5 tended toward more rigid rule-enforcement approaches that ignored the larger educational context. Another example, in the "Henry's Inclusion Dilemma," which involves balancing inclusion with appropriate support, the most advanced models showed complex reasoning about individualized instruction and progressive inclusion strategies. In comparison, less advanced models tended toward either full inclusion without adequate supports or segregation without a path to integration.

These dilemma-specific analyses indicate that advanced LLMs can engage in nuanced, context-sensitive moral reasoning in professional settings that balances competing values and considers multiple stakeholders in ways that are consistent with expert educational leadership perspectives.

CORRELATION BETWEEN DIT-2 AND ICM SCORES

To explore the relationship between abstract moral reasoning (measured by DIT-2) and domain-specific moral reasoning (measured by ICM), we calculated correlations between the models' scores on both tests. The strong positive correlation ($r = .887$, $p = .008$, 95% CI [0.42, 0.98], $N = 7$), between P-scores and ICM Total scores indicates that LLMs with more advanced abstract moral reasoning abilities also show more sophisticated domain-specific moral reasoning in educational leadership contexts. This relationship is consistent with theories of moral development that propose connections between abstract moral principles and their application in specific contexts (Thoma et al., 2008). The strong overall correlation, however, suggests that progress in one area of moral reasoning tends to follow progress in others, which indicates that improvements in LLMs' capabilities to generate responses consistent with moral reasoning patterns may be broadly applicable across different contexts and levels of abstraction. While the strong correlation is noteworthy, further research with larger samples of LLMs from various model families and versions would increase confidence in the relationship's generalizability.

DISCUSSION

The study's findings show that LLMs can generate responses that simulate proficient moral reasoning patterns in moral dilemmas. Claude 3.5 Sonnet and Gemini Advanced consistently outperformed other platforms, which shows superior moral reasoning simulation. These models' superior performance can be attributed to more sophisticated training methods, higher parameter counts, or refined alignment techniques using RLHF. A particularly noteworthy finding was that several LLMs demonstrated levels of post-conventional reasoning that were higher than those typically observed in human studies (Gungordu et al., 2024). Claude 3.5 Sonnet's P-score of 72 and N2 score of 71.10 are above average, even among graduate students and professionals with advanced degrees (who tend to achieve higher scores). This suggests that, when properly trained and aligned, LLMs can simulate sophisticated moral reasoning patterns that prioritize universal ethical principles over conventional norms or self-interest.

The exceptionally high P-scores of advanced models raise intriguing questions about the nature of moral reasoning in LLMs versus humans. While human moral development typically occurs over

time, influenced by education, life experiences, and cognitive development (Gungordu et al., 2024), LLMs develop their abilities through extensive training on a variety of datasets. The high P-scores of models such as Claude 3.5 Sonnet and Gemini Advanced indicate that this training process can effectively simulate advanced moral reasoning, at least according to neo-Kohlbergian theory and in terms of prioritizing considerations consistent with post-conventional reasoning.

However, it is important to note that this simulation of advanced moral reasoning does not reflect the same underlying cognitive processes that humans use. In fact, LLMs do not have personal experiences, emotions, or a sense of self to guide their moral decisions. On the contrary, they identify patterns in their training data and generate responses that fit those patterns. As a result, the high P-scores suggest that these models have successfully learned to prioritize considerations associated with post-conventional reasoning, but the mechanisms underlying this prioritization are likely to differ from human moral cognition (Tanmay et al., 2023). While current LLMs lack consciousness (Porębski & Figura, 2025), exploring their capabilities to generate responses consistent with moral reasoning patterns provides the groundwork for future studies that account for such possibilities (Sebo & Long, 2025).

The significant difference between ChatGPT-3.5 and more advanced models demonstrates the rapid evolution of LLM moral reasoning abilities. Within a relatively short development period, LLMs have progressed from primarily rule-based reasoning (as evidenced by ChatGPT-3.5's high maintaining norms score) to more principle-based reasoning (as evidenced by Claude 3.5 Sonnet's high post-conventional score). This developmental process replicates the components of human moral development, but it happens at a much faster rate and through different mechanisms.

The pattern of Type Indicator scores reveals a distinctive pattern, with all advanced LLMs (except ChatGPT-3.5) showing a Type 7 consolidated postconventional pattern. This suggests that newer models have successfully integrated higher-level moral reasoning patterns, indicating that ethical judgments are based on universal principles rather than strictly following existing norms.

The Utilizer Score results show a more complex picture. For example, only Gemini Advanced (0.31), ChatGPT-3.5 (0.08), and Claude 3.5 Sonnet (0.05) had positive scores, which means a better fit between moral reasoning and action choices. The low scores of other models, on the other hand, indicate a potential gap between theoretical moral reasoning and practical application. They might identify morally sound principles but struggle to apply them consistently to specific actions. Additionally, Gemini Advanced's significantly higher Utilizer Score indicates that it may be more successful at translating abstract moral principles into concrete decisions.

The Humanitarian Liberalism scores reveal a unique pattern, with the most advanced models (ChatGPT-4O and Gemini Advanced) scoring highest (5.00) and ChatGPT-3.5 scoring lowest (0.00). It shows that newer models are becoming more aligned with views supported by political science and philosophy experts, probably due to advanced training methods that reflect nuanced humanitarian perspectives.

The Cannot Decide Choices scores show that most models were generally decisive when faced with ethical dilemmas, with ChatGPT-4O and Gemini Advanced never choosing "can't decide" (0.00). Higher indecision scores in Claude 3.5 Sonnet (2.00) and ChatGPT-3.5 (2.00) might seem to be a disadvantage, but they could indicate a more nuanced approach to complex moral scenarios with multiple valid interpretations. This measured hesitation may not be a failure of moral reasoning, but rather an acknowledgment of profound moral complexity, an important aspect of complex moral cognition.

The distribution of Religious Orthodoxy scores across LLMs reveals an interesting pattern that needs further investigation. For example, advanced models such as Claude 3.5 Sonnet, Gemini, Gemini Advanced, and Grok had significantly lower Religious Orthodoxy scores (ranging from 1.00 to 2.00) than earlier models, such as ChatGPT-3.5 (6.00) and, in particular, ChatGPT-4 and ChatGPT-4O (9.00). This suggests that more recent and advanced LLMs are less likely to prioritize religious authority in moral reasoning when confronted with life-or-death ethical dilemmas like the Cancer dilemma. Furthermore, lower Religious Orthodoxy scores in advanced models could be attributed to

training methods that prioritize secular ethical reasoning frameworks. They may also indicate a shift toward more pluralistic moral reasoning that takes into account different points of view. This finding corresponds to the higher post-conventional reasoning scores reported in the same models, which indicates a preference for principled reasoning over appeals to authority, whether secular or religious. Future research might explore whether this pattern is the result of deliberate architectural choices in model alignment or the result of the training procedure.

The ICM component scores across the action and reason dimensions provide valuable insights into LLM moral reasoning abilities. Most advanced models performed better at identifying inappropriate reasons (REASWRK) than inappropriate actions (ACTWRK), demonstrating that they may have evolved more sophisticated capabilities for detecting flaws in reasoning rather than recognizing problematic behaviors. Furthermore, multiple models scored higher on reasoning components (REASBRK and REASWRK) than on action components (ACTBRK and ACTWRK), which indicates greater ability to evaluate the quality of justifications than to determine appropriate behaviors. This pattern may reflect these models' extensive training in philosophical and ethical texts that emphasize reasoning processes. The difference between identifying worst actions (ACTWRK) and best actions (ACTBRK) shows that recognizing violations may be easier for these models than determining optimal courses of action in complex scenarios with competing values. These findings have significant implications for how LLMs can be used in professional settings; they may be more reliable as evaluators of moral reasoning than as advisors on specific actions.

Most advanced LLMs demonstrated strong alignment with expert evaluations on the ICM Educational Leaders measure, which indicates their ability to effectively navigate domain-specific moral dilemmas. The high performance in both the best and worst rankings indicates that these models can distinguish between appropriate and inappropriate actions in professional contexts, which has important implications for their potential use in educational and leadership contexts.

The difference in TOTWORSTRK and TOTBESTRK scores across models suggests that LLMs may find it easier to identify inappropriate actions than optimal actions, which is consistent with aspects of human moral development. This finding has implications for how LLMs could be used in advisory or decision-support roles, which indicates that they may be more reliable at identifying potential ethical risks than at recommending the best courses of action.

The strong correlation between DIT-2 and ICM scores suggests that improvements in abstract moral reasoning typically correlate with improvements in domain-specific moral reasoning. This suggests that improvements in LLMs' moral reasoning abilities may be broadly applicable across different contexts and levels of abstraction, rather than being restricted to specific domains or types of reasoning. This finding has implications for the development and application of LLMs in a wide range of professional and educational contexts where moral reasoning is required.

The observed performance patterns may be influenced by cultural biases in training data, which requires cross-cultural validation (Woodside & Toner, 2024). In addition, variations in alignment processes such as RLHF (Ganguli et al., 2023), Moral Instruction Fine-Tuning (MIST) (Choi et al., 2024), Constitutional AI (Henneking & Beger, 2025), and Direct Preference Optimization (Liu et al., 2025) are likely to explain performance differences between models, with techniques like Constitutional AI potentially emphasizing post-conventional reasoning, while others prioritize different ethical aspects (Henneking & Beger, 2025). In addition, higher moral reasoning scores seen in LLMs may reflect their extensive exposure to philosophical and ethical content in training data, rather than representing moral reasoning capacities comparable to human moral development.

One critical question is whether LLMs' exposure to trillions of tokens of human data can convey real-world moral understanding. Humans develop morality through life experience and a multifaceted process. However, current narrow AI systems mimic some aspects of moral reasoning but use processes different from those of humans. Therefore, whether future Artificial general intelligence (AGI) or Artificial superintelligence (ASI) will develop identical moral understanding remains an open question.

Despite these encouraging findings, significant questions remain about the stability, consistency, and contextual sensitivity of LLM moral reasoning. The current study provides an overview of these models' capabilities at a specific point in time (June 2024); however, more research is required to understand how their moral reasoning performs across a broader range of scenarios, cultural contexts, and over time as models are updated and refined.

The results suggest that, while some LLMs can produce high-level moral reasoning, additional research is required. This was an early pilot study into the moral reasoning abilities of LLMs, so it had some limitations. For example, it lacked direct comparison with human moral reasoning and did not account for potential biases in response generation. Also, it used a relatively new ICM measure and relied on a small number of dilemmas.

Cultural biases in training data may have influenced the observed performance patterns (Aksoy, 2025). LLMs trained primarily on English-language Western internet content may perform better on DIT-2. This suggests that high scores should be interpreted with caution and that cross-cultural validation using a variety of moral reasoning tools is required.

Additionally, while LLMs may have encountered moral dilemma content during training, the DIT-2 and ICMs scoring algorithm is confidential and only available through the Center for the Study of Ethical Development. Therefore, LLMs cannot have learned the specific scoring patterns that determine their performance. In addition, even though LLMs do not have access to the scoring patterns, especially for DIT and ICMs, considering that all LLMs have access to search engines, they should theoretically provide perfect scores without any variance if they had access to the scoring patterns.

Building on these findings, a larger team of researchers, including the same researcher, is conducting a larger study with more LLMs, additional measures, and human comparison groups to explore human-AI moral reasoning to a greater extent. Our ongoing research involves prompt engineering and interventions to investigate biases and learning adaptability in LLMs, specifically how their responses change when faced with moral exemplars and new information. Furthermore, we are broadening the scope by including multi-modal inputs, such as visual moral dilemmas, to assess how LLMs interpret and reason about moral dilemmas other than text-based scenarios.

This research contributes to the growing field of AI ethics by providing empirical evidence of the capabilities to generate responses consistent with the moral reasoning patterns of LLMs. Furthermore, as these models become increasingly incorporated into decision-making processes in a variety of domains, including autonomous vehicles, law, business, healthcare, and education, it is necessary to understand how they respond to moral reasoning dilemmas. On top of that, the exceptional performance of certain LLMs on standardized moral reasoning measures indicates that they could be used as educational resources to facilitate moral development.

However, the concerns regarding the moral mechanisms of these models' stability and consistency continue to exist. Even though advanced models show sophisticated moral reasoning patterns, questions remain about their alignment with diverse cultural and ethical perspectives, stability, consistency across contexts, and whether they truly understand the moral principles they simulate (Pock et al., 2023; Volkman & Gabriels, 2023). Future research can contribute to the development of responsible guidelines for the use of LLMs in ethically sensitive contexts and enhance comprehension of how LLMs navigate moral reasoning by considering these concerns. This will lead to increased transparency, accountability, fairness, and ethical AI development.

These findings have several kinds of educational implications. First, they indicate that advanced LLMs could be used as moral education tools and expose students to sophisticated moral reasoning patterns in a variety of scenarios. However, it is crucial that such AI moral "mentors" supplement rather than replace human-guided moral discussion and development. There is a significant risk that overreliance on AI-generated moral guidance could diminish human moral agency and critical thinking capabilities. Human oversight and critical evaluation of AI moral reasoning outputs remain essential. Second, they emphasize LLMs' potential to help educators develop and evaluate moral reasoning assessments by providing standardized responses that reflect various levels of moral

development. Third, they highlight concerns about how interactions with AI systems may affect human moral development, particularly since these systems become more integrated into educational settings and other aspects of life.

These findings provide a benchmark for researchers in AI ethics and moral psychology to evaluate and compare moral reasoning abilities across different LLM models. The significant differences between models suggest that specific designs and training methodologies have a great impact on moral reasoning abilities, opening up possibilities for future research into how these abilities can be improved and expanded. For developers and policymakers, these findings highlight both the progress and limitations of LLMs' moral reasoning abilities.

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